A growing epithelial spheroid in an extracellular matrix: the Plexus2 implementation

`prototype/ecm` · runs in `log/okuda_ECM` · 6 August 2026

1. Sets, operators, schedule

A Plexus model is sets × fields × operators × schedule. The biological hierarchy here is four entities and one field; the basement membrane is a *specialised* ECM, not the ECM:

entity / set	representation	principal operators
epithelium `vertex`, `cell`	3D active vertex model	`seed_mesh_3d` build the shell `cell_geometry_3d` area, perimeter, volume per cell `shape_energy_3d` the mechanics `morphogen_growth_3d` grow each cell's target `divide_3d` divide on volume doubling `ecm_growth_gate_3d` matrix stress slows the cycle [7]
junction a relation, not a set	half-edges of the mesh, state keyed by vertex pair	`junction_myosin` per-junction myosin `reconnect_t1_3d` neighbour exchange (T1)
basement membrane `basement_membrane_particle`	nodes + explicit crosslinks	`seed_basement_membrane` lay the sheet down `basement_membrane_bond` the crosslink force `basement_membrane_remodel` rest lengths creep `basement_membrane_bond_break` fragmentation `basement_membrane_secrete` add material as it grows `integrin_adhesion` tether to the surface
surface `surface`	one node per membrane node: direction, radius	`surface_track` write the recorded boundary
interstitial ECM `mpm_particle`	MPM fibres	`seed_ecm` fibres, alignment, density `cell_to_ecm` the epithelium's push `cell_exclude_3d` non-penetration backstop `ecm_stress` colour by von Mises stress
(field) `mpm_grid`	MLS-MPM background grid	`mpm_scatter` / `mpm_grid_update` / `mpm_gather` the solve

2. The spheroid: 3D active vertex model

Cells are faces of a closed triangulated shell; cell f's volume is the wedge from the tissue centre, $v_f = \frac{1}{3}\,\mathbf{c}_f \cdot \mathbf{N}_f$. The energy is

$$E \;=\; \sum_f \Big[K_A\,(A_f - A_f^0)^2 + K_P\,(P_f - P_f^0)^2 + \tfrac{1}{2}\Gamma P_f^2 + K_V\,(v_f - v_f^0)^2\Big] \;+\; \Lambda \sum_e m_e\,\ell_e \;+\; K_R \sum_i \big(|\mathbf{x}_i| - R_0\big)^2, \quad (1)$$

relaxed overdamped each frame, $\mathbf{x} \leftarrow \mathbf{x} - \eta\mu\,\nabla E$ with a per-step cap. The factor m_e is new (§6); with $m_e \equiv 1$ Eq. (1) is the stock Okuda energy and every prior run is bit-identical. Growth scales each cell's targets by a cumulative s_f,

$$s_f \leftarrow s_f\big(1 + \lambda(\rho + \mathrm{Hill}(a_f))\big), \qquad A_f^0 = A_f^{0,\mathrm{init}} s_f^2, \quad P_f^0 = P_f^{0,\mathrm{init}} s_f, \quad v_f^0 = v_f^{0,\mathrm{init}} s_f^3, \quad (2)$$

and `divide_3d` splits a cell once v_f passes twice its reference. From 200 cells the tissue reaches ~ 6000 and its apical radius $4.66 \to 16.6$ (arbitrary units) in 402 frames.

In the vertex model every cell–cell contact pulls with the same strength. Real contractility is not uniform: myosin accumulates on some junctions and not others, and where it accumulates the junction pulls harder and shortens, which is how cells exchange neighbours. Giving each junction its own myosin is what lets that happen.

Each junction carries an amount of myosin m_e multiplying the line tension, chasing a target on its own timescale:

$$\Lambda \sum_e \ell_e \;\longrightarrow\; \Lambda \sum_e m_e\,\ell_e, \qquad m_e^{ss} = a\Big(1 + \beta\,\big(x_e/\langle x\rangle - 1\big)\Big), \qquad \dot m_e = \big(m_e^{ss} - m_e\big)/\tau_m. \quad (3)$$

What matters is what x_e is. If it is the junction's *length*, the loop reads *longer* → *more myosin* → *shorter*, which evens the cells out and suppresses neighbour exchange. Myosin is in fact recruited by *tension* [6], and accumulates on junctions in the act of disappearing — the opposite: a junction that pulls harder shortens, which makes it pull

Figure 1: Two cells share an edge, and that edge *is* the junction — it belongs to neither, which is why its state is keyed by the pair of vertices rather than stored on a cell. Myosin sits on the shared edge and multiplies its tension. Right: the feedback, which is the only thing here the original model cannot already do.

harder still. So x_e is the edge tension, $\Lambda m_e + 2K_P(P_f - P_f^0) + \Gamma P_f$, taken from the same energy the mechanics minimises.

The two versions cannot be told apart by the spread of junction lengths, because in the fast-myosin limit — the correct one, since myosin turns over in seconds — substituting the target into Eq. (3) turns the length version into a line tension plus a harmonic spring on each edge, and narrowing that spread is what such a spring does by definition. They differ in how often cells swap neighbours, so that is the observable we report: keyed to tension, the rate of neighbour exchange is roughly double.

Two things the model does not have. The myosin responds to the *magnitude* of tension and has no preferred direction in the plane of the sheet, so it cannot express the planar polarity that drives directional tissue elongation. And τ_m is a chosen number, not a measured one.

3. The ECM: MLS-MPM

The matrix is a material, not a set of springs, so it is solved the way materials are: as a cloud of points that carry the material and a background grid that does the physics. Each frame the points hand their mass and momentum to the grid nodes nearest them, the grid resolves everything arriving at each node into one velocity, and the points read that velocity back and move. Nothing is ever computed between two points directly — which is why bodies that never touch can still push on one another, and why the grid spacing sets the finest thing the matrix can feel.

We seed it as fibres rather than a uniform block, because a fibrous matrix and a homogeneous one respond differently to being pushed, and the interesting behaviour — splaying, dragging, a stress front travelling outward — lives in the fibres.

4. The basement membrane

A basement membrane is a thin sheet of protein an epithelium lays down underneath itself and sits on. It has to do two things at once that pull against each other: hold together as a sheet, and keep covering a surface that grows. We model the load-bearing part of it — the crosslinked collagen IV network — as nodes joined by springs. Three rules. A crosslink is a spring that slowly *forgets* being stretched and snaps if stretched too far; a node is tied to a fixed *direction* on the surface, which is what makes it adhesion rather than contact, since a node free to slide to the nearest point would never be stretched at all; and everything moves at a speed set by force against drag, with no mass:

$$\gamma\,\dot{\mathbf{x}}_i = \sum_{j\sim i} \mathbf{f}_{ij} \;+\; \mathbf{f}_i^{\mathrm{adh}}, \qquad \mathbf{f}_{ij} = k_b\,(L_{ij} - \ell_{ij}^0)\,\hat{\mathbf{d}}_{ij}, \qquad \dot{\ell}_{ij}^0 = (L_{ij} - \ell_{ij}^0)/\tau_r, \tag{4}$$

$$\mathbf{f}_i^{\mathrm{adh}} = \kappa\big(\mathbf{a}_i - \mathbf{x}_i\big), \qquad \mathbf{a}_i = \mathbf{c} + \hat{\mathbf{u}}_i\big(R(\hat{\mathbf{u}}_i, t) + \delta\big), \qquad \hat{\mathbf{u}}_i \text{ frozen at seeding,} \tag{5}$$

with a crosslink dying once $(L - \ell^0)/\ell^0 > \varepsilon_b$.

Forgetting is not optional: a sheet that only stretches cannot enclose a surface that more than triples in size. And because area grows as the square of the radius while new material is added as a fraction of what is already there, the sheet must be *secreted* as the tissue grows or it thins.

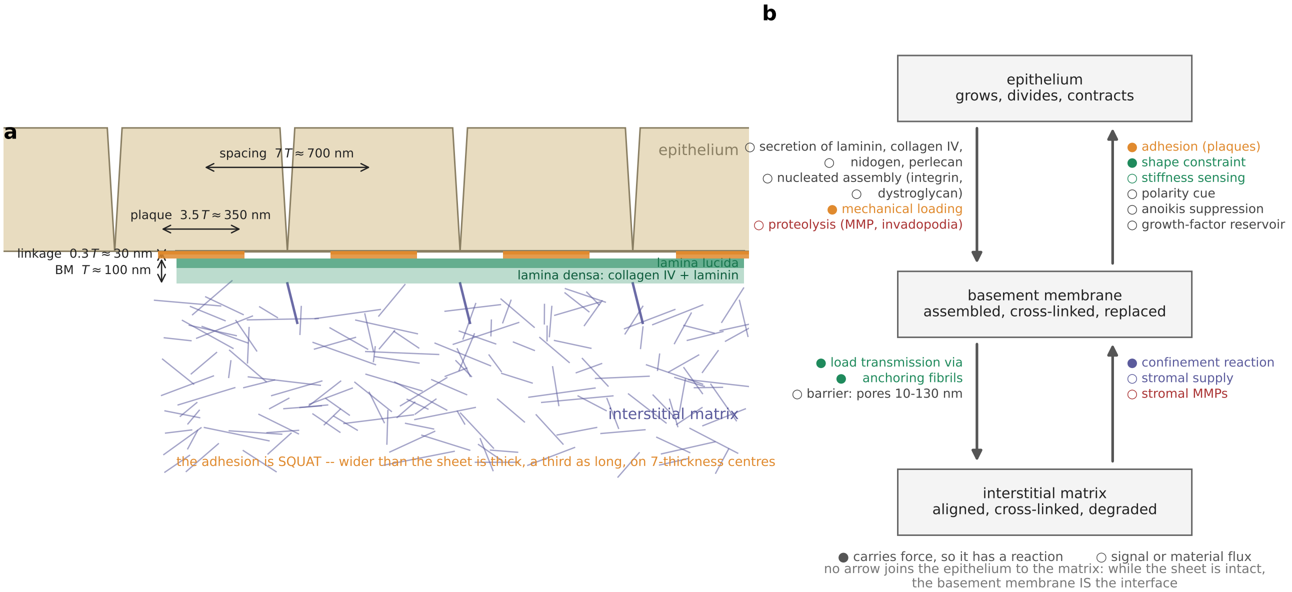

Figure 2: (a) The three entities in section, at the proportions the literature measures rather than the ones that are convenient: the adhesion is a squat plaque, not a thread, and the sheet is thin against its own spacing. (b) Every operator, in the direction it acts. Filled bullets carry a force and must therefore have a reaction; open bullets carry a signal or a material flux. There is no arrow between the epithelium and the matrix, because while the sheet is intact the basement membrane *is* the interface — an epithelium that also pushes the stroma directly is counting the same push twice.

Two consequences of having no mass. Only the ratio k_b/γ appears in Eq. (4) — a relaxation *rate* — so a stiffness on its own has no meaning here, and the question worth asking is whether the sheet relaxes faster than the tissue grows. The same ratio bounds what a fixed time step can carry, $\Delta t\,(z\,k_b + \kappa)/\gamma < 2$ for z crosslinks meeting at a node, which is about twenty times the bound an inertial sheet would have. Raising γ moves it further, at the price of a sheet that responds more slowly.

Finally, what is called adhesion here is a tether to a surface, not a receptor: the dominant anchor in a real epithelium is to laminin, which is not modelled, and there is no attachment to the surrounding stroma either, so this tether is the only thing holding the sheet.

5. Spheroid ↔ BM ↔ ECM

Three objects, and every arrow between them is a measured mechanism rather than a modelling choice. The first thing the measurements say is that the middle object is not a passive spacer. A basement membrane is ∼100 nm thick, its adhesions are *squat* — plaques ∼350 nm across on ∼700 nm centres, joined to the cell across a ∼30 nm gap [18], so an adhesion is three times wider than the sheet is thick and a third as long — and the sheet is 10 to 100 times stiffer than the stroma outside it, not equal to it. The second thing they say is that it is not a permanent solid: collagen IV in the fly embryo has a half-life of 7–10 hours [17] and in mouse hair follicle about 3, and slowing that turnover disrupts morphogenesis. A sheet enclosing a tissue that triples its radius does not stretch to 2.4; it is *replaced* at the new size while it is being stretched.

Mass, not just mechanics. The sheet carries an areal density ρ of collagen IV that is produced by the epithelium and removed by proteolysis, on a surface whose area $A(t)$ is growing:

$$\frac{\mathrm{D}\rho}{\mathrm{D}t} = \underbrace{s(\mathbf{x},t)}_{\text{secretion}} - \underbrace{\frac{\rho}{\tau_{\mathrm{bm}}}}_{\text{turnover}} - \underbrace{\rho\,\frac{\dot{A}}{A}}_{\text{dilution by growth}}, \qquad \tau_{\mathrm{bm}} = \frac{t_{1/2}}{\ln 2} \approx 4\text{–}14\ \text{h}. \tag{6}$$

Homeostasis is $s^* = \rho\,(1/\tau_{\mathrm{bm}} + \dot{A}/A)$, and the two terms are comparable here: a spheroid tripling its radius over a day has $\dot{A}/A \sim 10^{-1}\,\mathrm{h}^{-1}$ against $1/\tau_{\mathrm{bm}} \sim 10^{-1}\,\mathrm{h}^{-1}$. Turnover is not a correction to growth; it is the same size as growth, which is why the sheet can accommodate it at all.

A nonlinear sheet. A single Young's modulus is wrong by construction: the same basement membrane reads ∼0.4 MPa at low strain and ∼3 MPa at high strain [16], and a growing spheroid takes it across that whole range. The cheapest law that respects it makes the tangent modulus rise with the principal stretch $\lambda = \sigma_{\max}(\mathbf{F})$,

$$E(\lambda) = E_0\big[1 + \beta\,(\lambda - 1)\big], \qquad E_0 \approx 0.4 \text{ MPa}, \quad \beta \approx 5, \qquad \frac{E_{\rm bm}}{E_{\rm ecm}} \in [10, 100]. \tag{7}$$

Adhesion with its reaction, and with slip. An integrin plaque is a two-sided object: whatever force it applies to the sheet it applies equally and oppositely to the cell, which a replayed epithelium currently discards. And the sheet is not pinned — it *slides* relative to the cells — so the tangential law is friction against relative velocity, not a spring to a frozen direction:

$$\mathbf{f}_p^{\rm adh} = \kappa_n\,\big(\ell_p - \ell_0\big)\,\hat{\mathbf{n}}_p \;-\; \xi\,\big(\mathbf{v}_{\rm bm} - \mathbf{v}_{\rm epi}\big)_{\parallel}, \qquad\qquad \mathbf{f}_p^{\rm epi} = -\,\mathbf{f}_p^{\rm adh}, \tag{8}$$

$$n_p = \Sigma^{-2} \text{ plaques per unit area}, \quad \Sigma \approx 7\,T, \qquad\qquad \ell_0 \approx 0.3\,T. \tag{9}$$

ℓ_0, the linkage's rest length, is what sets the standoff — a material property, not a tuning constant — and ξ is what lets a sheet stay attached while the tissue slides underneath it.

Sheet to stroma. The two are joined by anchoring fibrils [19], so the stroma is loaded through the sheet and never directly by the cells; the same fibrils are what let a stiff sheet spread a local epithelial push over a soft matrix. With $\mathbf{t}$ the traction the sheet exerts on the stroma,

$$\mathbf{t} = k_a\,\big(\mathbf{x}_{\rm ecm} - \mathbf{x}_{\rm bm}\big) \quad \text{on anchored points}, \qquad \text{and} \;\; \mathbf{f}^{\rm bm} = -\sum_{\rm anchors} \mathbf{t}, \tag{10}$$

with the barrier being geometric rather than mechanical: pores of 10–130 nm pass solutes, stop cells, and are what a proteolytic operator has to open before anything crosses.

What this asks of the implementation. Four things, and each is a separate operator with its own test: a mass balance on the sheet (Eq. 6) rather than a fixed particle budget; a strain-dependent modulus (Eq. 7) rather than one number; an adhesion that returns its reaction and permits slip (Eq. 8) rather than a tether to a frozen direction; and an anchorage to the stroma (Eq. 10) that removes the epithelium's direct push on it.

6. Plexus2 container

Two questions decide whether the biology above can be written down: what the *sets* are, and what kind of operator each mechanism is. Both have answers the literature forces, and both differ from what the archived line built.

Sets. An adhesion is not a body and a fibril is not a particle — each is a *relation between two entities*, which in Plexus2 is an edge-set carrying `.pre` and `.post` index buffers rather than a set of nodes with positions. Writing the hemidesmosome as an edge-set is what makes its reaction automatic: one operator on `plaque` returns a delta to both endpoints, so the force the sheet feels and the force the cell feels cannot get out of step. The archived line made the adhesion first a field on the membrane and then a column of MPM particles, and both lost the reaction.

set	kind	one element is	state it carries
`spheroid`	node, depth 0	the organ	centroid, cell count
`cell`	node, depth 1	one epithelial cell	A, P, V and their targets, age
`vertex`	node, depth 1	a corner of the cell mesh	`pos`, `vel`
`junction`	**edge** $\subseteq$ `vertex`^{2}	a cell–cell contact	length, myosin
`bm`	node, depth 1	a patch of basement membrane	`pos`, $\mathbf{F}$, ρ (areal density), age since deposition
`plaque`	**edge** `cell`$\to$`bm`	one hemidesmosome	rest length ℓ_0, load, bound flag
`ecm`	node, depth 1	a patch of interstitial matrix	`pos`, $\mathbf{F}$, fibre direction
`fibril`	**edge** `bm`$\to$`ecm`	one anchoring fibril	rest length, load
`mpm_grid`	field	momentum on a background lattice	mass, momentum
`protease`	field	MMP concentration	scalar, diffusing
`growth_factor`	field	FGF/VEGF bound by perlecan	scalar, bound + free

Operators. One row per mechanism in Fig. 3b, in Plexus2's own taxonomy: *Lateral* moves state along an edge set, *Exchange* scatters to or gathers from a field, *Rewire* changes which elements are connected, and *Divide*/*Die* change how many elements exist. The last column is what would falsify the operator — the measurement its test folder has to make.

operator	kind	acts on	mechanism, from the literature	what its test measures
epithelium				
`grow_3d`, `divide_3d`	Divide	`cell`	volume growth and cytokinesis [3, 4]	cells 200 $\rightarrow$ 6076, radius $\times 3.56$
`shape_energy_3d`	Lateral	`junction`	vertex-model force balance [5]	aspect 1.021, stays round
`reconnect_t1_3d`	Rewire	`junction`	neighbour exchange [4]	0.0089 T1 per cell per frame
epithelium $\rightarrow$ basement membrane				
`bm_secrete`	Divide	`bm`	polarised deposition of laminin, collagen IV, nidogen, perlecan; the sheet turns over on a half-life of hours [17, 15]	areal density ρ held constant as $A(t)$ grows
`bm_assemble`	Rewire	`bm`	laminin polymerises where integrin and dystroglycan nucleate it [15]	no sheet forms where the receptor is absent
`bm_degrade`	Die	`bm`	MMP proteolysis, local and MMP-independent breach [20]	a hole opens only where protease is present
basement membrane $\leftrightarrow$ epithelium				
`plaque_seed`	Rewire	`plaque`	hemidesmosomes form at density Σ^{-2}, $\Sigma \approx 7T$ [18]	plaque count and spacing
`plaque_pull`	Lateral	`plaque`	Eq. (8): normal spring at rest length ℓ_0, tangential friction ξ, *and its reaction* [18]	standoff $= \ell_0$; momentum conserved to machine precision
`plaque_rupture`	Rewire	`plaque`	the bond fails past a load	fraction bound vs. applied stress
`bm_stiffness`	Lateral	`bm`	Eq. (7): tangent modulus rising with stretch [16, 13]	E 0.4 $\rightarrow$ 3 MPa across the run
basement membrane $\leftrightarrow$ matrix				
`fibril_seed`, `fibril_pull`	Rewire, Lateral	`fibril`	collagen VII anchoring [19]; the stroma is loaded *through* the sheet	stroma displacement with the sheet present vs. absent
`mpm_scatter/gather`	Exchange	`bm`, `ecm`	both bodies share one momentum field [1, 2]	contact without an explicit contact law
`bm_barrier`	Exchange	`bm`, fields	pores 10–130 nm pass solutes, stop cells [15]	solute flux across an intact vs. breached sheet
matrix				
`ecm_align`, `ecm_crosslink`	Lateral	`ecm`	fibres rotate toward the maximum principal stretch, which is what carries force beyond one cell diameter [11, 12]	alignment angle vs. radial direction

Three of these have no counterpart in the archived implementation and are the reason it stalled: `bm_secrete` as a mass balance rather than a fixed particle budget, `plaque_pull` returning its reaction, and `fibril_pull` replacing the epithelium's direct push on the stroma. Each gets its own folder, spec, movie and measurement plot, in the order of the table.

11. Symbols

Grouped by what they belong to. Where a symbol is a stiffness, the second column says what it is a stiffness *of*, since several are easy to confuse.

Epithelium — Eqs. (1), (2)		
A_f, P_f, V_f	area, perimeter, volume	of cell (face) f; A_f^0, P_f^0, V_f^0 are its targets
K_A, K_P, K_V	stiffnesses	how hard a cell resists changing its area / perimeter / volume
Λ	line tension	pull along every cell–cell contact, one value for the whole tissue
Γ	perimeter contractility	a second, quadratic pull on each cell's perimeter
ℓ_e	length of junction e	
K_R, R_0	radial restoring stiffness, radius	holds the shell against collapse
f_{grow}	per-frame growth factor	applied to a cell's target volume
Junction myosin — Eq. (3)		
m_e	myosin on junction e	dimensionless multiplier on Λ; mean 1
x_e	what the feedback reads	the junction's tension
a	global myosin level	multiplies every junction equally
β	feedback strength	how strongly m_e responds to x_e above average
τ_m	myosin timescale	how fast m_e chases its target
Surface and matrix — Eqs. (**??**), (**??**)		
$\hat{\mathbf{u}}_i$	direction	fixed unit vector from the tissue centre $\mathbf{c}$
$R(\hat{\mathbf{u}}, t)$	surface radius	how far the epithelium reached in that direction
r_i	radius of particle i	
k_c	contact stiffness	how hard the surface pushes a fibre back out
P	matrix pressure	contact force per unit surface area, by direction
$p_{1/2}, n, f$	gate half-point, sharpness, floor	g is the resulting growth multiplier, $f \leq g \leq 1$
Basement membrane — Eqs. (4), (5)		
$\mathbf{x}_i$	position of node i	
γ	drag	converts force to speed: $\dot{\mathbf{x}} = \mathbf{F}/\gamma$
k_b	**crosslink stiffness**	how hard a stretched crosslink pulls its two nodes together
L_{ij}, ℓ_{ij}^0	current, rest length	of the crosslink between nodes i and j
$\hat{\mathbf{d}}_{ij}$	unit vector	from node i toward node j
τ_r	remodelling timescale	how fast ℓ^0 creeps toward L, i.e. how fast it forgets
ε_b	break strain	the stretch at which a crosslink snaps
κ	**tether stiffness**	how hard a node is pulled back to its anchor (not k_b)
$\mathbf{a}_i$	anchor	the point on the surface node i is tied to
δ	offset	how far outside the surface the sheet sits

12. References

The operators of §10 are grouped here by what they claim. Where a row of that table states a number — a half-life, a plaque spacing, a modulus — the citation is the measurement it was taken from, not a review that mentions it.

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